Project Planning

Phase 4: Project Planning Phase

1. Planning Objective

The project plan organizes work into stages so that the main user journey can be built and tested before optional enhancements are attempted. The highest priority is a complete vertical slice: enter preferences, generate an itinerary, display it, save it and retrieve it. Additional features should be added only after this core path works reliably.

2. Work Breakdown

Requirements and scope

Define the target user, primary problem, required trip fields, expected itinerary structure and boundaries. Confirm that booking and verified live place data are not part of the first implementation.

Design and architecture

Separate the frontend, API routes, AI service, schemas and database client. Define the request and response structure before connecting the frontend to the backend. Decide how the application will communicate errors and show progress.

Frontend development

Build the trip form, generation state, itinerary display and saved-trip controls. Add the Hotels & Restaurants section and browser print flow after the primary itinerary workflow is functional.

Backend development

Implement health, itinerary, saved-trip and place-search routes. Configure the AI provider and Supabase client through environment variables. Validate request and response structures and handle external-service failures.

Integration and testing

Run the frontend and backend locally, verify the API base URL, generate a realistic itinerary and test persistence. Check deletion, map-search links and printing. Record observed outcomes instead of assuming that code presence means a test passed.

Deployment and documentation

Deploy the frontend and backend separately, perform a deployed smoke test, capture application screenshots and complete the documentation. Record the demo video after the application and report have been reviewed.

3. Milestones

1. Scope and requirements approved.
2. Architecture and data contracts defined.
3. Core trip form and AI generation implemented.
4. Supabase persistence and Saved Trips integrated.
5. Hotels & Restaurants and print/save actions integrated.
6. Frontend and backend deployed.
7. Functional tests and screenshot evidence collected.
8. Eight documentation phases completed and demonstration recorded.

4. Risk Management

AI service availability

Model quotas, rate limits and network failures can interrupt generation. The application should display a clear error and allow a user to retry when appropriate.

Database configuration

Missing credentials or schema mismatch can prevent saving. Verify environment variables and database schema in the deployment environment without exposing secrets.

Misleading travel information

Generated costs and activities may be inaccurate. Label costs as estimates and require users to verify real-world details. Avoid describing Google Maps links as verified inventory.

Incomplete submission evidence

Screenshots, test logs and demo video are separate deliverables. Assign time to collect them and verify that screenshots do not reveal keys or personal tokens.

5. Completion Criteria

The project is ready for final review when the core flow works in the deployed environment, all stated limitations are documented, screenshots are inserted with captions, test outcomes reflect actual runs, and the demonstration link is accessible to the intended reviewers.